

Full Stack Development with MERN

Project Documentation – Flight Finder

1. Introduction

- Project Title: Flightfinder: Navigating Your Air Travel Options
- Team ID: LTVIP2026TMIDS80455
- Team Size: 5
- Team Leader: Harika Kapireddygari
- Team Member: Amidala Veeresh
- Team Member: Lakkineni Balaram
- Team Member: Mamilla Prudhvinath Goud
- Team Member: Kandavaram Manoj Kumar
- Repository: <https://github.com/kapireddygariharika/FlightFinder.git>

2. Project Overview

- Purpose: The purpose of this project is to build a full-stack flight booking application using the MERN stack as part of the SmartInternz internship.
- Features:
 - User registration and login
 - Search and book flights
 - Admin dashboard
 - Responsive UI using React
 - REST APIs with MongoDB backend

3. Architecture

- Frontend: Built using React.js with components for user interaction.
- Backend: Built using Node.js and Express.js for REST API functionality.
- Database: MongoDB for storing users, flights, and bookings.

4. Setup Instructions

- Prerequisites:
 - Node.js

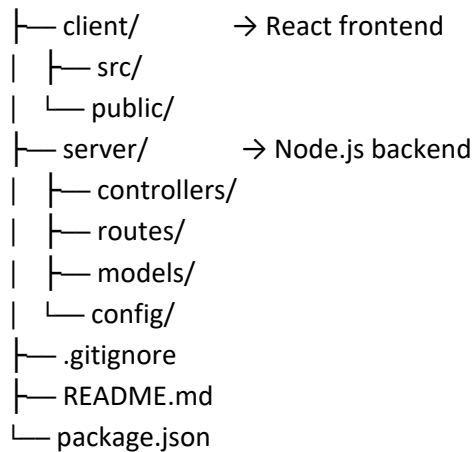
- MongoDB
- Git

- Installation:

1. Clone the repo: `git clone https://github.com/kapireddygariharika/FlightFinder.git`
2. `cd Flightfinder`
3. `cd client && npm install`
4. `cd ../server && npm install`
5. Create `.env` file with `MONGO_URI` and `JWT_SECRET`

5. Folder Structure

Flight-Finder/



6. Running the Application

- Frontend:
`npm start` (inside client folder)
- Backend:
`node index.js` (inside server folder)

7. API Documentation

- User APIs: `/register`, `/login`, `/my-bookings`
- Flight APIs: `/flights`, `/book`
- Admin APIs: `/admin/add-flight`, `/admin/delete-flight/:id`

8. Authentication

- JWT-based authentication with protected routes
- Token stored in local storage and verified via middleware

9. User Interface

- React with Bootstrap styling
- Components: Login/Register, Booking Form, Admin Panel

10. Testing

- Manual testing using Postman
- Console-based checks in frontend for validation

11. Screenshots or Demo

- Add screenshots from app (home, login, booking) here.
- Alternatively, provide demo link if deployed.

12. Known Issues

- No password reset feature
- No email confirmation on booking
- Admin route not login-restricted

13. Future Enhancements

- Add payment gateway
- Deploy app online
- Add filters and confirmation emails